

Solution

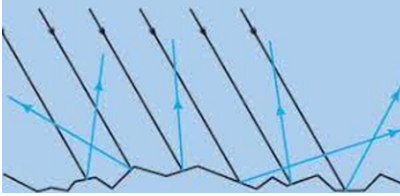
LIGHT-01

Class 08 - Science

Section A

1. Read the text carefully and answer the questions:

On the bases of law of reflection imagine that parallel rays are incident on an irregular surface. As we know the laws of reflection are valid at each point of the surface. Using law of reflection rays are constructed at various points.



- (i) (a) Wood

Explanation: Wood

- (ii) (a) regular reflection

Explanation: regular reflection

- (iii) (c) regular reflection

Explanation: regular reflection

- (iv) 1. irregular reflection

- (v) (b) False

Explanation: False

2. Read the text carefully and answer the questions:

An activity is going in the class by the teacher

Step 1 - Fix a white sheet of paper on a drawing board or a table.

Step 2 - Take comb and close all its openings except one in the middle.

Step 3 - Hold the comb perpendicular to the sheet of paper.

Step 4 - Throw light from a torch through the opening of the comb from one side

Step 5 - With slight adjustment of the torch and the comb you will see a ray of light along the paper on the other side of the comb.

Step 6 - Keep the comb and the torch steady. Place a strip of plane mirror in the path of the light ray.



- (i) (d) reflected ray

Explanation: reflected ray

- (ii) (c) equal

Explanation: equal

- (iii) (d) normal

Explanation: normal

- (iv) 1. reflection

- (v) (a) True

Explanation: True

3. Read the text carefully and answer the questions:

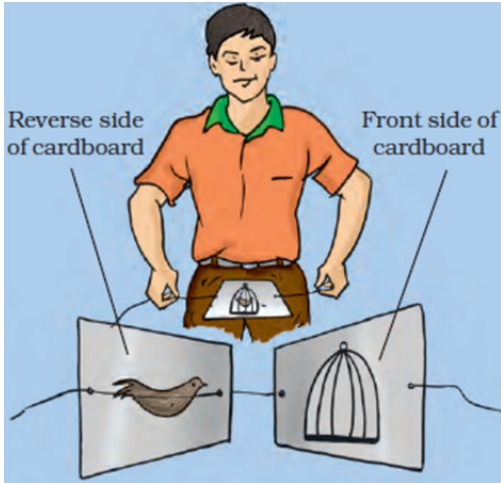
Step 1 - Get a square piece of cardboard of side 6-8 cm.

Step 2 - Make two holes.

Step 3 - Thread a string through the two holes.

Step 4 - Draw/ paste a cage on one side of the cardboard and a bird on the other side.

Step 5 - Twist the string and make the card twirl rapidly.



- (i) **(b)** $\frac{1}{16}$ second

Explanation: $\frac{1}{16}$ second

- (ii) **(b)** 25 cm

Explanation: 25 cm

- (iii) **(d)** Infinity

Explanation: Infinity

- (iv) 1. eyelids

- (v) **(a)** True

Explanation: True

4. Read the text carefully and answer the questions:

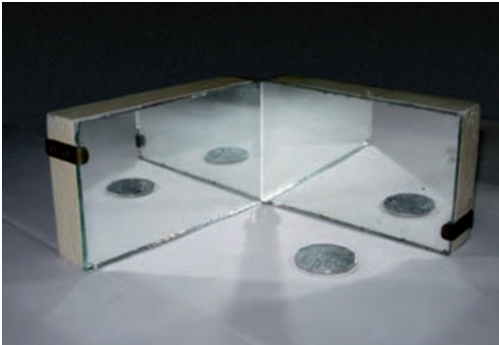
Step 1 - Take two plane mirrors. Set them at right angles to each other with their edges touching.

Step 2 - To hinge them you can use adhesive tape.

Step 3 - Place a coin in between the mirrors.

Step 4 - Now hinge the mirrors using the adhesive tape at different angles, say 45° , 60° , 120° etc.

Step 5 - Place some object (say a candle) in between them. Note down the number of images of the object in each case.



- (i) **(d)** 8

Explanation: 8

- (ii) **(b)** Kaleidoscope

Explanation: Kaleidoscope

- (iii) **(c)** different pattern

Explanation: different pattern

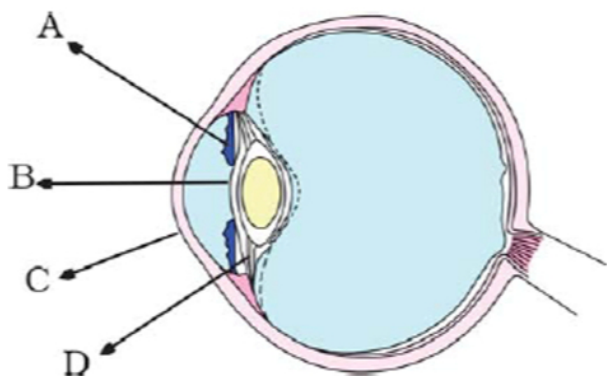
- (iv) **(a)** True

Explanation: True

- (v) 1. 30 cm

5. Read the text carefully and answer the questions:

We see things only when light coming from them enters our eyes. Eye is one of our most important sense organs. The eye has a roughly spherical shape. Outer coat of the eye is white. It is tough so that it can protect the interior of the eye from accidents.



- (i) **(b)** Iris

Explanation: Iris

- (ii) **(b)** both cones and rods

Explanation: both cones and rods

- (iii) **(c)** Cornea

Explanation: Cornea

- (iv) 1. retina

- (v) **(a)** True

Explanation: True

6. Read the text carefully and answer the questions:

Teacher of class 7 teaching the students about how rainbow formed for which she is conducting the following activity.

Take a glass prism. Allow a narrow beam of sunlight through a small hole in the window of a dark room to fall on one face of the prism. Let the light coming out of the other face of the prism fall on a white sheet of paper or on a white wall.



- (i) **(c)** all of these

Explanation: all of these

- (ii) **(d)** dispersion

Explanation: dispersion

- (iii) **(c)** seven colours

Explanation: seven colours

- (iv) 1. Rainbow

- (v) **(a)** True

Explanation: True

Section B

7. **(a)** Both A and R are true and R is the correct explanation of A.

Explanation: Periscope is used in submarine to see the objects in depth.

8.

- (d)** A is false but R is true.

Explanation: Lack of vitamin A causes eye disease. We should eat green vegetables like spinach broccoli etc.

9.

- (d)** A is false but R is true.

Explanation: Wood is not a smooth surface, so it will give diffused reflection.

10. **(a)** Both A and R are true and R is the correct explanation of A.

Explanation: In air, when sunlight shines on dust particles, they reflect the light and cause irregular reflection.

11. **(a)** Both A and R are true and R is the correct explanation of A.

Explanation: At barber's shop we can have back view of our hair which is due to multiple reflections.

12. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: When light incident along normal to plane mirror it will retrace its path and angle of incidence and Angle of reflection will be equal to zero degree.
13.
(d) A is false but R is true.
Explanation: When light is focused on back of eye on the layer of retina, it always forms a image but at the junction there is no vision. That spot is called blind spot.
14. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: There is a pupil in the iris and iris controls the amount of light enter into the eye.
15.
(b) Both A and R are true but R is not the correct explanation of A.
Explanation: Sunlight is called for white light which consists of seven colours.
16.
(c) A is true but R is false.
Explanation: Number of images formed depends upon the angle between two mirrors.
17.
(d) A is false but R is true.
Explanation: Braille system is used for visually impaired persons and people learn the words, letters and special character by touching the dots.
18. **(a)** Both A and R are true and R is the correct explanation of A.
Explanation: An owl has large cornea and large pupil to allow more light in the eye.
19.
(d) A is false but R is true.
Explanation: Our eye is spherical in shape and it shaped like a ball.

Section C

20. State True or False:

- (i) **(b)** False
Explanation: False
- (ii) **(b)** False
Explanation: False
- (iii) **(b)** False
Explanation: False
- (iv) **(b)** False
Explanation: False: Image formed in a plane mirror is always erect.
- (v) **(b)** False
Explanation: False
- (vi) **(a)** True
Explanation: True
- (vii) **(b)** False
Explanation: False
- (viii) **(b)** False
Explanation: False
- (ix) **(a)** True
Explanation: True
- (x) **(a)** True
Explanation: True
- (xi) **(b)** False
Explanation: False

(xii) (b) False

Explanation: False

(xiii) (b) False

Explanation: False

(xiv) (a) True

Explanation: True

(xv) (a) True

Explanation: True

(xvi) (a) True

Explanation: True

(xvii) (a) True

Explanation: True

(xviii) (b) False

Explanation: False

(xix) (b) False

Explanation: False: Plane mirror is used by barbers to show the back of head.

(xx) (a) True

Explanation: True

Section D

21. (a) A

Explanation: Following the laws of reflection of light, in case A, when a ray of light from the bulb strikes the mirror M, it will get reflected back such that the angle of incidence is equal to the angle of reflection.

22.

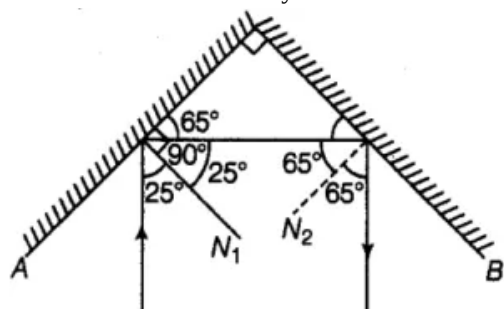
(c) All of these

Explanation: Laws of reflection are always followed whatever be the surface of the object. Therefore, the angle of incidence is equal to the angle of reflection on all three surfaces (S_1 , S_2 , and S_3).

23. (a) 50°

Explanation:

The Angle of reflection for the ray reflected from mirror B will be 65° because the figure below shows how reflected ray from mirror A forms incident ray on mirror B and then reflected back by an angle of 65° .



24. (a) C

Explanation: The transparent front part of the eye is cornea labelled as C (this is a thin layer).

25.

(d) A

Explanation: Eye should be placed at A because the hole can be seen only when the angle of incidence is equal to the angle of reflection.

Section E

26.	Column A	Column B
	(i) Cornea	(a) transparent front part of eye.
	(ii) Pupil	(b) is a small opening in the cornea

(iii) Iris	(c) controls the size of the pupil
(iv) Retina	(d) layer on which impression of images is formed
(v) Blind spot	(e) point on retina where there are no nerve endings
(vi) Rods	(f) sensitive for dim light
(vii) Cones	(g) sensitive for bright light

27. i. - b
ii. - c
iii. - d
iv. - a

28. i. - d
ii. - c
iii. - b
iv. - a

29.	Column A	Column B
	(i) Normal vision	(b) 25 cm
	(ii) Blind spot	(d) No sensory cells
	(iii) Pupil	(a) Controlled by Iris
	(iv) Braille system	(e) Visually challenged
	(v) Day light birds	(c) Kite, eagle
	(vi) rainbow	(f) Dispersion of light

30. i. - b
ii. - c
iii. - d
iv. - a

31. i. - d
ii. - c
iii. - a
iv. - b

32. i. - c
ii. - a
iii. - d
iv. - b